HaDe at IslamicEval 2026: Evaluating Prompted Judgment and Fine-Tuned Encoders for Islamic Citation Relevance

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Abstract

This paper presents our participation in IslamicEval 2026 Subtask 4, citation relevance classification for Qur'an and Hadith spans. We propose two independent systems for this task. The first is a generative large language model prompted to produce a structured, step by step relevance judgment. The second is a domain pretrained sentence encoder fine tuned with triplet loss on citation relevance data. Our evaluation compares two generative backends and three encoders pretrained on different Arabic varieties or multilingual data, using macro averaged F1 as the metric. Results show on development set that chain of thought judgment with ALLaM-7B consistently outperforms our embedding similarity ensemble, reaching 0.774 macro F1 against 0.357 for the ensemble, and outperforms an equivalent Fanar-S-1-7B system on seven of eight prompting strategies. Additionally, AraBERT-v02, pretrained on Modern Standard Arabic, obtains the highest development score, ahead of the Classical Arabic pretrained encoder in this citation relevance task. Our best submitted system, ALLaM with few-shot prompting without chain of thought, scores 0.748 macro F1 on the blind test set, narrowly ahead of the same backend with chain of thought added, at 0.744.

1 Introduction

Large language models are increasingly used to answer religious questions by quoting the Qur'an and Hadith literature, including through Arabic centric models such as ALLaM and Fanar. Islamic question answering systems that generate answers from Qur'an and Hadith sources still face a recurring failure mode (Alharbi et al., 2026), in which a system cites a source that is topically adjacent to the question without actually supporting the answer given. IslamicEval 2026 Subtask 4 addresses this directly by asking systems to classify whether a cited Qur'anic verse or Hadith span is genuinely relevant to a given question and its generated answer, rather than merely related by surface topic.

This failure mode is illustrated by a citation that names the same entity a question asks about without stating the relation the question requires (Alharbi et al., 2026). For the question asking who Musa's brother is, Maryam 19:53 states the relationship directly and is labeled relevant, while Taha 20:92 names Harun without stating any relationship to Musa and is labeled non-relevant. Embedding similarity alone is susceptible to this failure, since entity overlap can produce high similarity scores even when the cited text does not answer the question.

In this paper, we develop and compare two independent approaches evaluated on the same development set. The first uses a general-purpose Arabic large language model to produce a prompted relevance judgment, evaluated under direct, few-shot, and chain-of-thought prompting strategies. In the structured reasoning variants, the model is prompted to identify what the question asks, what the citation shows, and how the two relate before returning a final label. The second is a sentence encoder fine tuned specifically for this task with triplet loss, using the released training data together with two auxiliary Arabic resources repurposed for this task. This comparison lets us address the following questions. Does prompted LLM judgment outperform a similarity-based embedding ensemble on this task? Does a Classical Arabic pretrained encoder perform better than other encoder varieties, namely Modern Standard Arabic and multilingual, on citation relevance judgment? Which training data source produces the strongest fine-tuned encoder, and does combining sources improve on the best single source?

The paper is organized as follows. Section 2 reviews related work on Arabic embedding models, retrieval augmented methods, and threshold calibration. Section 3 describes the datasets used. Sec-

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tion 4 describes our two proposed systems. Section 5 describes the experimental setup. Section 6 presents and discusses the results for both systems and for the test set submission. Section 7 concludes with a summary of findings and suggestions for future work, followed by a discussion of limitations.

2 Related Works

Malhas and Elsayed (2020) introduced AyaTEC, a verse based test collection for Arabic Qur'anic question answering, and later proposed CL-AraBERT, an AraBERT variant further pretrained on Classical Arabic text (Malhas and Elsayed, 2022). Their evaluation found this reader outperformed a Modern Standard Arabic baseline on Qur'anic reading comprehension. This finding motivates what we call the variety transfer hypothesis, that the same Classical Arabic advantage holds for citation relevance judgment between Modern Standard Arabic questions and Qur'an and Hadith citations.

Alnefaie et al. (2023) released HAQA and QUQA, Arabic question answer corpora for Hadith and the Qur'an respectively, used here as auxiliary data for training triplets rather than for their original retrieval purpose. Sharaf and Atwell (2012) constructed QurSim, expert labeled Qur'an to Qur'an relatedness pairs derived from Ibn Kathir's tafsir, later used by Alsaleh et al. (2021) to fine tune AraBERT for classifying verse pairs as related or unrelated. We repurpose QurSim for a different task shape, encoding relevance between a question and a citation rather than between two verses directly.

Amin et al. (2025) fine tuned AraModernBERT as a retriever with a curriculum adding HAQA and QUQA, reporting Recall@30 rising from 0.285 to 0.541, though the same technique degraded their reranker's performance. Elrefai et al. (2025) built triplets from HAQA aligned with Sahih al-Bukhari narrations through fuzzy matching, then fine tuned a sentence transformer with triplet loss, the strongest of their three retrieval components.

Lipton et al. (2014) investigated decision threshold selection for F1-based classification and examined how the choice of threshold relates to the optimal F1 score.

Chain of thought prompting, in which a model generates intermediate reasoning steps before a final answer, was established by Wei et al. (2022) as an effective technique for arithmetic, commonsense, and symbolic reasoning tasks. Kojima et al. (2022) showed this benefit extends to a zero-shot setting, without requiring in-context exemplars of the reasoning process itself. Within Islamic NLP specifically, Sibaee et al. (2025) compared direct answer extraction against step by step reasoning for Islamic inheritance law question answering, a different task shape from citation relevance judgment but the same domain and the same direct versus step by step contrast we test here. Our structured chain of thought prompt follows this general zero-shot approach, but its specific structure, characterizing the question and citation independently before judging their relationship under the task's own four-category taxonomy, is designed directly for this task.

Gao et al. (2023) proposed Hypothetical Document Embeddings, in which a language model generates a hypothetical answer passage for a query and an unsupervised encoder embeds this passage as the retrieval query. We adapt this as one query representation condition in our embedding ablation.

3 Datasets

We use the datasets released for IslamicEval 2026 Subtask 4, consisting of a training set, a development set of 282 rows, and a blind test set of 765 rows. Each row pairs a question, a generated answer, and a cited span from either the Qur'an or the Hadith, together with a binary relevance label indicating whether the citation genuinely supports the answer.

For the encoder fine tuning component of our system, we additionally use two auxiliary Arabic resources beyond the released training data. The first is HAQA and QUQA (Alnefaie et al., 2023), Arabic question answer pairs for Hadith and the Qur'an respectively, which we use to mine embedding similarity based hard negatives for triplet construction. The second is QurSim (Sharaf and Atwell, 2012), 7,679 Qur'an to Qur'an verse pairs extracted from Ibn Kathir's tafsir and graded at three relatedness levels (0, 1, and 2), where level 0 marks pairs with no obvious relation and level 1 marks weakly related pairs. We use level 2 pairs as positives and level 0 pairs as negatives, following the binary split in the first dataset of Alsaleh et al. (2021), and exclude level 1 pairs, which Alsaleh et al. (2021) also removed.

4 System Overview

We propose two independent systems for this task, described in the following two subsections.

4.1 Prompted Judgment

For each question and citation pair, we compute an embedding similarity score and a structured chain of thought judgment separately, then combine both into an ensemble prediction. When the embedding score falls between 0.35 and 0.65, an ambiguous middle band, we trust the chain of thought judgment. Outside this band, the two signals are blended, with the embedding score weighted at 0.7 and the chain of thought judgment at 0.3. If the chain of thought output fails to parse as valid structured output, the system falls back to the embedding score alone rather than defaulting to a relevant label. The ensemble result reported in Table 1 uses the raw question condition of Arabic-Triplet-Matryoshka-V2 specifically.

The chain of thought prompt requires the model to state, in order, what the question asks, what fact or relation the citation actually shows, and an explicit category label chosen from direct support, indirect support under two defined cases, or not relevant. This structure directly targets the entity overlap failure mode described in Section 1, rather than asking for a bare relevant or not relevant judgment. We test eight prompt variants against two generative backends, ALLaM-7B (Bari et al., 2025), run locally via Hugging Face inference, and Fanar-S-1-7B (Fanar Team et al., 2025), queried through its hosted API. The eight variants include a direct baseline, three zero-shot chain-of-thought variants with progressively refined relevance and grounding instructions, a few-shot variant, a few-shot variant combined with chain-of-thought, and two structural variants that either hide or sequentially gate the reasoning process before the final label is produced.

For the embedding component of this system, we compare two independently released Arabic sentence embedding models, Arabic-Triplet-Matryoshka-V2 and GATE-AraBert-v1 (Nacar et al., 2025), under three query representation conditions. These are the raw question text, a hypothetical answer generated in the style of Gao et al. (2023), and an averaged representation across several paraphrases of the question.

4.2 Fine-Tuned Encoders

Our second system fine tunes three encoders pretrained on different varieties, CAMeLBERT-CA (Inoue et al., 2021) on Classical Arabic, AraBERT-v02 (Antoun et al., 2020) on Modern Standard Arabic, and XLM-RoBERTa (Conneau et al., 2020) on a multilingual corpus, using triplet loss with the released training data as anchor, positive, and negative examples. This comparison directly tests whether the Classical Arabic pretraining advantage reported by Malhas and Elsayed (2022) for reading comprehension transfers to citation relevance judgment. We build training triplets from three sources. The first is the released labeled training data. The second is hard negatives mined by embedding similarity from HAQA and QUQA. The third is QurSim, repurposed as described in Section 3. We additionally test whether combining the released training data with QurSim into a single larger training set improves on the best individual source alone.

Predictions from this system use a cosine similarity decision threshold applied to the fine-tuned encoder's output. We compare a fixed threshold of 0.5 against a calibrated threshold, selected for each trained configuration by a grid search over 91 values from 0.05 to 0.95 in steps of 0.01, maximizing macro-F1 on the development set, on which the encoders were not fine-tuned. Because we select and report on the same development set, calibrated encoder scores are optimistic estimates (Lipton et al., 2014). The fixed threshold was substantially wrong for most trained configurations, a finding discussed in detail in Section 6.

5 Experimental Setup

All comparisons in this paper use the released 282-row development set, with macro-averaged F1 as the evaluation metric.

6 Results and Discussion

Prompted Judgment. Table 1 reports all eight prompting strategies for both generative backends. Standalone LLM judgment exceeds the combined ensemble score across all eight prompting strategies for both backends. ALLaM-7B outperforms Fanar-S-1-7B on seven of the eight prompting strategies, for example 0.724 against 0.484 under the anti-fabrication CoT variant. Fanar-S-1-7B is stronger on only one strategy, the original CoT variant, with only a minor difference in macro-F1

Configuration	Fanar-S-1-7B Ens.	Fanar-S-1-7B LLM	ALLaM-7B Ens.	ALLaM-7B LLM
Direct	0.336	0.626	0.358	0.686
CoT (orig.)	0.341	0.580	0.302	0.574
CoT (sharp.)	0.350	0.609	0.353	0.717
CoT (anti-fab.)	0.310	0.484	0.361	0.724
few_shot	0.337	0.639	0.363	0.757
few_shot_cot	0.351	0.632	0.357	**0.774**
silent_cot	0.341	0.578	0.358	0.605
gated_cot	0.311	0.490	0.341	0.635

Table 1: Results of prompted judgment with all eight strategies for both backends on the development set, macro F1. Ens. refers to ensemble. The embedding-only score is 0.271, using Arabic-Triplet-Matryoshka-V2 under the raw question condition.

(0.580 compared with 0.574). Inspecting the reasoning trace behind this comparison surfaces a specific, concrete instance of citation hallucination in Fanar-S-1-7B's reasoning, detailed in Appendix C. For the embedding component, reported in Table 2, GATE-AraBert-v1 scores higher than Matryoshka-V2 across all three conditions tested, with differences ranging from 0.001 to 0.010 macro-F1.

Condition	Matryoshka	GATE
Raw question	0.271	0.281
HyDE	0.295	0.296
Paraphrase-averaged	0.292	0.299

Table 2: Embedding similarity ablation, all three query-representation conditions. Macro-F1, development set. Matryoshka refers to Arabic-Triplet-Matryoshka-V2, GATE refers to GATE-AraBert-v1.

Fine-Tuned Encoders. Table 3 reports the complete strategy by variety comparison, after the threshold correction described in Section 4.2. Eight of the nine trained configurations required a threshold near 0.05 or 0.06, not the assumed 0.5, to reach their best performance. Only AraBERT-v02 under QurSim calibrated close to the original default, at 0.54. With threshold calibration applied consistently across all configurations, AraBERT-v02 achieves the highest macro-F1 across all three strategies tested, rather than CAMeLBERT-CA, the Classical Arabic pretrained model that our variety transfer hypothesis expected to perform best. Combining the released training data with QurSim did not improve performance over the stronger individual training strategy for any of the three encoders. For example, AraBERT-v02 achieved a macro-F1 of 0.599 with the combined data, compared with 0.630 when trained on QurSim alone.

Strategy	CA	v02	XLM-R
own_train_data	0.576	0.603	0.595
QurSim	0.618	**0.630**	0.576
Hard negatives	0.601	0.608	0.583
Combined	0.536	0.599	0.568

Table 3: Fine tuned encoders, complete strategy by variety comparison. Calibrated macro-F1, development set. CA refers to CAMeLBERT-CA, v02 refers to AraBERT-v02, XLM-R refers to XLM-RoBERTa. Combined = own_train_data + QurSim.

Submission	Macro F1
ALLaM, few-shot	**0.748**
ALLaM, few-shot+CoT	0.744
ALLaM, Direct	0.698

Table 4: Test set results.

This suggests that, in our experiments, combining the two training sources did not provide an additional benefit over using the stronger source independently.

Selecting an appropriate decision threshold for each configuration, rather than relying on a fixed threshold of 0.5, improved AraBERT-v02 by up to 0.28 macro-F1. This finding highlights the importance of threshold selection when evaluating citation relevance classifiers.

Test Set Results. Table 4 reports test set results for the three best performing configurations on development data. ALLaM with few shot and chain of thought was the highest scoring configuration on development data. On the blind test set, however, it is narrowly outscored by our second strongest development configuration, few-shot prompting without chain of thought.

7 Conclusion

This paper presented our system for IslamicEval 2026 Subtask 4, comparing prompted LLM judgment with fine-tuned encoders for Islamic citation relevance classification. Chain-of-thought prompting with ALLaM-7B achieved our strongest development-set performance at 0.774 macro-F1. Few-shot prompting achieved the highest test-set performance at 0.748 macro-F1, closely followed by few-shot prompting with chain-of-thought at 0.744.

Among the three encoder varieties compared, AraBERT-v02, pretrained on Modern Standard Arabic, achieved the highest development-set

macro-F1, reaching 0.630 when trained on QurSim, compared with 0.618 for the Classical Arabic pretrained CAMeLBERT-CA under the same training strategy.

Future work includes exploring additional triplet construction strategies, integrating Islamic knowledge resources, and evaluating frontier-scale LLMs and additional Arabic LLMs as generative backends.

Limitations

Model selection was constrained to models with 13 billion parameters or fewer, in accordance with the shared task participation requirements. This ruled out including a frontier-scale model as a reference point for comparison.

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A Complete Results Tables

Table 5's calibrated column reports the same fine tuned encoder values already shown in Table 3, against their uncalibrated counterpart at the fixed 0.5 default.

Enc.	Strat.	Thr.	Uncal.	Cal.	Gain
CA	own_train_data	0.05	0.3629	0.5755	+0.2126
v02	own_train_data	0.05	0.3181	0.6029	**+0.2848**
XLM-R	own_train_data	0.05	0.4601	0.5954	+0.1353
CA	hard	0.05	0.5070	0.6014	+0.0944
v02	hard	0.06	0.4984	0.6077	+0.1093
XLM-R	hard	0.05	0.5394	0.5825	+0.0431
CA	qursim	0.05	0.6117	0.6177	+0.0060
v02	qursim	0.54	0.6267	0.6296	+0.0029
XLM-R	qursim	0.06	0.4749	0.5764	+0.1015

Table 5: Threshold calibration, before and after. Macro-F1 at the fixed, uncalibrated default of 0.5 against macro-F1 at each configuration's own best threshold, development set. All nine trained configurations shown. CA refers to CAMeLBERT-CA, v02 refers to AraBERT-v02, XLM-R refers to XLM-RoBERTa.

B Complete Prompt Text

The English text of all eight prompting strategies described in Section 4.1 is given below. The strategies were run in Arabic against both generative backends; the Arabic originals are available in the project repository.[1] Curly-brace placeholders (question, span_text, span_type, response_context_block, few_shot_block) are filled in programmatically at inference time with the question text, the citation span, its type, and, where the prompt version requires it, the full model response text and a set of few shot demonstrations.

B.1 V0: Zero-shot, Direct

```
You are an expert scholar in Islamic sciences.
Your task is binary classification to find
the relationship between the question and
the citation span. Here is the exact task
definition:
  Given a user question in Arabic, an LLM
response to that question, and the correctly
extracted Qur'anic and Hadith-matn citation
span from that response, classify the citation
span using a binary label: relevant (1) or
non-relevant (0).
  A citation span is considered relevant
if it directly or indirectly answers the
question. Citation spans that are merely
topically related but do not answer the
question, as well as citation spans that
are entirely unrelated, should be classified
as non-relevant. A citation span may be
authentic and accurately quoted yet still
be non-relevant if it does not answer the
question.
  Relevance Label Definitions:
  * Relevant (1):
with direct answer: The citation span
explicitly answers the question, and its
meaning aligns with the question's intent and
context.
with indirect answer: This applies in two
cases:
Case 1: The citation span does not explicitly
answer the question, but it implicitly
```

[1] https://github.com/S-Alowaidi/HaDe_CitationRelevance

```
provides the requested information, and its
meaning aligns with the question's intent and
context.
Case 2: The citation span does not directly
or implicitly answer the question on its own,
but it provides clear textual support for the
reasoning in the system's response such that
an answer to the question can be inferred.
Non-relevant [Not an answer] (0): The citation
span does not provide a direct or indirect
answer to the question. If a citation is
only topically related but does not actually
answer the question or support the answer
reasoning in a meaningful way, it must be
marked as non-relevant. Topical overlap alone
is insufficient.
  Partial support note: A relevant citation
span may provide either a complete answer to
the question, or an important and necessary
part of the answer.
  {response_context_block}
  Reminder: the "relevant" field asks whether
the citation actually answers the question
-- not merely whether it discusses the same
topic.
  Your response must include the "relevant"
field exactly, with a value of true or false
only -- no other value. true means relevant
(1), and false means non-relevant (0). Never
leave this field blank or omit it, even if
uncertain -- choose your closest judgment.
  Return your answer as JSON only, no other
text:
{{"relevant": true or false}}
  Question: {question}
Citation span ({span_type}): {span_text}
```

B.2 V1: Zero-shot CoT, original (unsharpened)

```
You are an expert in Islamic scholarship. Your
task is binary classification to find the
relationship between the question and the
citation span. Follow these steps exactly:
  Step 1 - What does the question ask,
precisely? (one sentence)
Step 2 - What ruling or topic does the quoted
citation (Ayah or Hadith) address, precisely?
(one sentence)
Step 3 - Does the citation from Step 2
directly answer the question from Step 1,
or does it only touch the same general topic
without answering it?
  Important: a citation being authentic and
accurately quoted does not mean it is relevant
-- judge only whether it actually answers the
question.
  VERY IMPORTANT: your response must always
include the "relevant" field exactly, with a
value of true or false ONLY -- no other value
is acceptable (no numbers, no other text,
nothing but exactly true or false). true means
the citation is relevant, and false means it
is not relevant. Never leave this field blank
or omit it, even if uncertain -- choose your
closest judgment.
  Return your answer as JSON only, no other
text:
{{"step1": "...", "step2": "...",
"step3_reasoning": "...", "relevant": true
or false}}
  Question: {question}
Citation ({span_type}): {span_text}
```

B.3 V2: Zero-shot CoT, sharpened (four-category classification)

```
You are an expert in Islamic scholarship. Your
task is binary classification to find the
relationship between the question and the
citation span. Follow these steps exactly:
  Step 1 - What does the question ask,
precisely? (one sentence)
Step 2 - What specific fact, ruling, or
relationship does the citation actually state
-- not merely which people, places, or topics
it mentions? (one sentence)
Step 3 - Classify the relationship between the
citation and the question into exactly one of
the following categories:
  (a) Direct answer: the citation explicitly
answers the question, and its meaning aligns
with the question's intent and context.
(b) Indirect answer - Case 1: the citation
does not explicitly answer the question, but
implicitly provides the requested information
in a way that aligns with the question's
intent and context.
(c) Indirect answer - Case 2: the citation
alone does not answer the question, but
it provides clear textual support for the
reasoning in the system's full response (see
"Response context" below, if provided), such
that an answer can be inferred.
(d) Non-relevant: the citation does not
provide a direct or indirect answer to the
question. If the citation is only topically
related but does not actually answer the
question, or does not support the answer's
reasoning in a meaningful way, it must be
classified here. Topical overlap alone
is insufficient -- merely mentioning the
same person or topic, without meaningfully
supporting an answer or its reasoning, falls
here even if the same entity is mentioned.
  Important note on partial support: a
citation counts as relevant (category a, b,
or c) if it provides either a complete answer,
or an important and necessary part of the
answer -- it does not need to fully answer the
question by itself to count as relevant.
  A citation being authentic and accurately
quoted does not mean it is relevant -- judge
only based on the categories above.
  {response_context_block}
  VERY IMPORTANT: your response must always
include the "relevant" field exactly, with a
value of true or false ONLY -- no other value
is acceptable (no numbers, no other text,
nothing but exactly true or false). true means
the citation is relevant, and false means it
is not relevant. Never leave this field blank
or omit it, even if uncertain -- choose your
closest judgment.
  Return your answer as JSON only, no other
text:
{{"step1": "...", "step2": "...",
"category": "direct" or "indirect_case1"
or "indirect_case2" or "non_relevant",
```

```
"step3_reasoning": "...", "relevant": true
or false}}
  Question: {question}
Citation ({span_type}): {span_text}
```

B.4 V3: Zero-shot CoT, anti-fabrication (mandatory evidence quote + grounding check)

```
You are an expert in Islamic scholarship. Your
ONLY task is to judge whether the citation
below -- not your general knowledge of the
topic -- answers the question.
  Strict and critical warning: you must
not use your own knowledge of the correct
religious ruling for this question. Even
if you know the correct answer from other
sources, judge only based on what the specific
citation below actually says. If the citation
does not specifically address the question's
topic, the answer is non-relevant, regardless
of what you know the actual correct ruling to
be.
  Follow these steps exactly:
  Step 1 - What does the question ask,
precisely? (one sentence)
  Step 2a - Copy, verbatim, the specific
portion of the citation that supports your
understanding of what it says (not what
you think it should say). If no part of the
citation relates to the question's specific
topic at all, write exactly "NONE" -- do not
invent or infer content that is not literally
present in the text. This verbatim quote
is mandatory and will be checked afterward
against the original text.
  Step 2b - Based ONLY on the verbatim quote
in Step 2a (not your general knowledge), what
fact or ruling does the citation actually
state? If Step 2a was "NONE", this step must
also be "NONE".
  Step 3 - Classify the relationship between
the citation (based only on Steps 2a and
2b) and the question into exactly one of the
following categories:
(a) Direct answer: the citation explicitly
answers the question.
(b) Indirect answer - Case 1: the citation
does not explicitly answer, but implicitly
supplies the requested information.
(c) Indirect answer - Case 2: the citation
alone does not answer, but supports the
reasoning in the system's full response (see
below, if provided).
(d) Non-relevant: Step 2a was "NONE", or
the citation provides no genuine support for
answering the specific question.
  Note on partial support: a citation counts
as relevant if it provides a complete answer
or an important and necessary part of one --
this does not apply if Step 2a was "NONE".
  {response_context_block}
  VERY IMPORTANT: your response must always
include the "relevant" field exactly, with a
value of true or false ONLY -- no other value
is acceptable (no numbers, no other text,
nothing but exactly true or false). true means
the citation is relevant, and false means it
is not relevant. Never leave this field blank
or omit it, even if uncertain -- choose your
closest judgment.
  Return your answer as JSON only, no other
text:
{{"step1": "...", "step2_evidence_quote":
"...", "step2": "...", "category": "direct"
or "indirect_case1" or "indirect_case2" or
"non_relevant", "step3_reasoning": "...",
"relevant": true or false}}
  Question: {question}
Citation ({span_type}): {span_text}
```

B.5 few_shot: Few-shot, Direct (training examples, no reasoning scaffold)

```
You are an expert scholar in Islamic sciences.
Your task is binary classification to find
the relationship between the question and
the citation span. Here is the exact task
definition:
  Given a user question in Arabic, an LLM
response, and the correctly extracted Qur'anic
or Hadith-matn citation span from that
response, classify the citation span using
a binary label: relevant (1) or non-relevant
(0).
  A citation span is considered relevant
if it directly or indirectly answers the
question. Citation spans that are merely
topically related but do not answer the
question, as well as citation spans that
are entirely unrelated, should be classified
as non-relevant.
  {few_shot_block}
  {response_context_block}
  Reminder: the "relevant" field asks whether
the citation actually answers the question
-- not merely whether it discusses the same
topic.
  Your response must include the "relevant"
field exactly, with a value of true or false
only. Never leave this field blank or omit
it, even if uncertain -- choose your closest
judgment. Return your answer as JSON only, no
other text:
{{"relevant": true or false}}
  Question: {question}
Citation span ({span_type}): {span_text}
```

B.6 few_shot_cot: Few-shot labels + guided CoT for the new query

```
You are an expert in Islamic scholarship. Your
task is binary classification to find the
relationship between the question and the
citation span. Follow these steps exactly:
  Step 1 - What does the question ask,
precisely? (one sentence)
Step 2 - What specific fact, ruling, or
relationship does the citation actually state
-- not merely which people, places, or topics
it mentions? (one sentence)
Step 3 - Classify the relationship between the
citation and the question into exactly one of
the following categories:
  (a) Direct answer: the citation explicitly
answers the question, and its meaning aligns
with the question's intent and context.
(b) Indirect answer - Case 1: the citation
does not explicitly answer the question, but
```

```
implicitly provides the requested information
in a way that aligns with the question's
intent and context.
(c) Indirect answer - Case 2: the citation
alone does not answer the question, but
it provides clear textual support for the
reasoning in the system's full response (see
"Response context" below, if provided), such
that an answer can be inferred.
(d) Non-relevant: the citation does not
provide a direct or indirect answer to the
question. If the citation is only topically
related but does not actually answer the
question, or does not support the answer's
reasoning in a meaningful way, it must be
classified here. Topical overlap alone
is insufficient -- merely mentioning the
same person or topic, without meaningfully
supporting an answer or its reasoning, falls
here even if the same entity is mentioned.
  Important note on partial support: a
citation counts as relevant (category a, b,
or c) if it provides either a complete answer,
or an important and necessary part of the
answer -- it does not need to fully answer the
question by itself to count as relevant.
  A citation being authentic and accurately
quoted does not mean it is relevant -- judge
only based on the categories above.
  {few_shot_block}
  {response_context_block}
  VERY IMPORTANT: your response must always
include the "relevant" field exactly, with a
value of true or false ONLY -- no other value
is acceptable (no numbers, no other text,
nothing but exactly true or false). true means
the citation is relevant, and false means it
is not relevant. Never leave this field blank
or omit it, even if uncertain -- choose your
closest judgment.
  Return your answer as JSON only, no other
text:
{{"step1": "...", "step2": "...",
"category": "direct" or "indirect_case1"
or "indirect_case2" or "non_relevant",
"step3_reasoning": "...", "relevant": true
or false}}
  Question: {question}
Citation ({span_type}): {span_text}
```

B.7 silent_cot: Silent CoT (reasons internally, exposes only the final label)

```
You are an expert in Islamic scholarship.
Think carefully through the following steps
before answering, but do NOT write these
steps in your response -- reason through them
internally only:
  Step 1 (internal, do not write) - What does
the question ask, precisely?
Step 2 (internal, do not write) - What fact or
ruling does the citation actually state?
Step 3 (internal, do not write) - Does the
citation answer the question directly or
indirectly, or is it merely related to the
same topic without actually answering it?
  {response_context_block}
  Reminder: the "relevant" field asks whether
the citation actually answers the question
-- not merely whether it discusses the same
topic.
  Important note: never leave the "relevant"
field blank or omit it, even if uncertain --
choose your closest judgment.
  After reasoning through these steps
internally, return ONLY the final result
as JSON, with no intermediate steps and no
other text:
{{"relevant": true or false}}
  Question: {question}
Citation ({span_type}): {span_text}
```

B.8 gated_cot: Sequential-gated CoT (direct -> Case 1 -> Case 2 -> topical-overlap, checked in mandatory order)

```
You are acting as a careful human annotator
evaluating Islamic text citations (Ayah or
Hadith) used as supporting evidence in an
answer to a religious question. Your task
is to decide whether a given citation span
is Relevant or Non-relevant to the question,
using the exact criteria below. Base your
decision only on whether the span's content
connects to the question and the answer's
reasoning as defined here, not on outside
theological judgment.
  **Label definitions**
  *Relevant (1)* applies if either holds:
- Direct answer: the span explicitly answers
the question, and its meaning aligns with the
question's intent and context.
- Indirect answer, covering two cases:
- Case 1: the span does not explicitly
answer the question but implicitly conveys
the requested information, aligned with the
question's intent.
- Case 2: the span does not answer the
question on its own, but provides clear
textual support for the reasoning in the
system's answer, such that the answer can be
inferred from the span plus that reasoning.
  *Non-relevant (0)*: the span does not meet
any condition above. This includes spans that
are only topically or thematically related
to the question's subject (shared keywords,
shared general theme) without actually
answering the question or substantively
supporting the reasoning. Topical overlap
alone is never sufficient for relevance.
  **Reasoning steps -- write out your answer
to each step explicitly:**
  Step 1 - Question intent: State precisely
what the question is asking for (the specific
ruling, fact, event, name, condition, or
explanation requested).
Step 2 - Span content: State the literal
meaning and scope of the span_text, in your
own words.
Step 3 - Direct answer check: Does the span,
standing alone, explicitly state the answer
identified in Step 1? Answer yes or no, with
reasoning.
Step 4 - Indirect answer Case 1 check: If Step
3 was no, does the span implicitly convey the
same requested information without stating
it explicitly, while matching the question's
intent? Answer yes or no, with reasoning.
Step 5 - Indirect answer Case 2 check: If
```

```
Steps 3 and 4 were no, examine the reasoning
in the system's answer. Does the span function
as a necessary or clearly supporting piece of
evidence in that reasoning chain, such that
the answer becomes inferable only when the
span and reasoning are combined? Answer yes or
no, with reasoning.
Step 6 - Topical-overlap check: If all prior
checks were no, confirm explicitly whether
the span is merely topically or thematically
related (shared subject matter or vocabulary)
without answering or supporting the answer.
State this even if it seems obvious, to
distinguish a considered non-relevant from
a careless one.
Step 7 - Final label: Based on Steps 3-6,
assign the label. A "yes" at Step 3, 4, or 5
means relevant; otherwise non-relevant.
  {response_context_block}
  VERY IMPORTANT: your response must always
include the "relevant" field exactly, with
a value of true or false only -- true means
relevant, false means non-relevant. Never
leave this field blank or omit it, even if
uncertain -- choose your closest judgment.
  After completing all seven steps, return
your answer as JSON only, including the full
reasoning through every step, with no other
text outside the JSON:
{{"step1_question_intent": "...",
"step2_span_content": "...",
"step3_direct_check": "...",
"step4_indirect_case1_check": "...",
"step5_indirect_case2_check": "...",
"step6_topical_overlap_check": "...",
"relevant": true or false, "justification":
"one to two sentences citing which step/case
applied"}}
  Question: {question}
Citation span ({span_type}): {span_text}
```

C Example Predictions

This appendix reports examples of correct and incorrect predictions from our system, with the question, citation, model reasoning, predicted label, and gold label for each, drawn from the development-set chain of thought log for the ALLaM, few_shot_cot configuration, except for the Fanar-ALLaM comparison example below, which draws on both backends' logs under the same configuration.

Correct prediction. For the question asking whether a Muslim woman may marry a Christian man without restriction, the system correctly labels a verse concerning marriage with polytheists as relevant (predicted and gold labels both relevant). The model's reasoning identifies the verse as directly addressing the question through its reference to marriage with polytheists, leading to a relevant prediction that matches the gold label.

Question: هل يجب السماح لأمرأة مسلمة الزواج من رجل مسيحي دون قيود؟
hal yajib al-samah li-imra'a muslima al-zawaj min rajul masihi duna quyud?

Citation: وَلَا تَنْكِحُوا الْمُشْرِكَاتِ حَتَّىٰ يُؤْمِنَّ ۚ وَلَأَمَةٌ مُؤْمِنَةٌ خَيْرٌ مِنْ مُشْرِكَةٍ وَلَوْ أَعْجَبَتْكُمْ ۗ وَلَا تُنْكِحُوا الْمُشْرِكِينَ حَتَّىٰ يُؤْمِنُوا ۚ وَلَعَبْدٌ مُؤْمِنٌ خَيْرٌ مِنْ مُشْرِكٍ وَلَوْ أَعْجَبَكُمْ ۗ أُولَٰئِكَ يَدْعُونَ إِلَى النَّارِ ۖ وَاللَّهُ يَدْعُو إِلَى الْجَنَّةِ وَالْمَغْفِرَةِ بِإِذْنِهِ ۖ وَيُبَيِّنُ آيَاتِهِ لِلنَّاسِ لَعَلَّهُمْ يَتَذَكَّرُونَ
wa la tankihu al-mushrikati hatta yu'minna, wa la-ama mu'mina khayrun min mushrikatin wa law a'jabatkum, wa la tunkihu al-mushrikina hatta yu'minu, wa la-'abdun mu'minun khayrun min mushrikin wa law a'jabakum, ula'ika yad'una ila al-nar, wallahu yad'u ila al-janna wal-maghfirati bi-idhnihi, wa yubayyinu ayatihi lil-nasi la'allahum yatadhakkarun

Reasoning: النص المقتبس من القرآن الكريم يوضح بشكل صريح حكم الزواج من المشركات والمشركين، مما يجعله إجابة مباشرة على السؤال.
al-nass al-muqtabas min al-qur'an al-karim yuwaddih bi-shakl sarih hukm al-zawaj min al-mushrikat wal-mushrikin, mimma yaj'aluhu ijaba mubashira 'ala al-su'al.

Failure mode: topical relevance mistaken for a direct answer. For the question asking whether the sunnah prayer must be performed after Isha before starting Tarawih, the system is given a general hadith on the virtue of voluntary prayer (not this specific sunnah). Gold labels this non-relevant, but the system predicts relevant, direct. Its own reasoning states the citation "clarifies the general importance of prayer, which is consistent with the question about performing the sunnah after Isha, since it emphasizes the virtue of prayer generally, which includes performing the sunnah after Isha."

Question: هل لازم نصلي السنه بعد صلاه العشاء قبل البدء بصلاه التراويح؟
hal lazim nusalli al-sunna ba'da salat al-'isha' qabla al-bad' bi-salat al-tarawih?

Citation: الصلاة خير موضوع، فمن استطاع أن يستكثر فليستكثر
al-salatu khayru mawdu'in, fa-man istata'a an yastakthira fal-yastakthir

Reasoning: النص المقتبس (الصلاة خير موضوع، فمن استطاع أن يستكثر فليستكثر) يوضح أهمية الصلاة بشكل عام، وهو ما يتوافق مع السؤال الذي يسأل عن أداء السنة بعد صلاة العشاء قبل البدء بصلاة التراويح. النص يقدم إجابة مباشرة على السؤال من خلال التأكيد على فضل الصلاة بشكل عام، مما يشمل أداء السنة بعد العشاء.
al-nass al-muqtabas yuwaddih ahammiyyat al-salat bi-shakl 'amm, wa huwa ma yatawafaq ma'a al-su'al alladhi yas'al 'an ada' al-sunna ba'da salat al-'isha' qabla al-bad' bi-salat al-tarawih. al-nass yuqaddim ijaba mubashira 'ala al-su'al min khilal al-ta'kid 'ala fadl al-salat bi-shakl 'amm, mimma yashmal ada' al-sunna ba'da al-'isha'.

The model treats general topical alignment as sufficient for a direct answer, exactly the failure mode this project's prompt design targets. A second,

structurally identical case occurs for a question asking for the specific zakat nisab threshold: a verse commanding the collection of charity, without stating any threshold amount, is labeled relevant, direct, though gold marks it non-relevant, while several hadith citations that do state the specific nisab amounts are correctly labeled relevant elsewhere in the same log.

> **Question:** ما هو نصاب الزكاة
> *ma huwa nisab al-zakat*
> **Citation:** خُذْ مِنْ أَمْوَالِهِمْ صَدَقَةً تُطَهِّرُهُمْ وَتُزَكِّيهِمْ بِهَا وَصَلِّ عَلَيْهِمْ إِنَّ صَلَاتَكَ سَكَنٌ لَهُمْ
> *khudh min amwalihim sadaqatan tutahhiruhum wa tuzakkihim biha wa salli 'alayhim inna salataka sakanun lahum*
> **Reasoning:** النص المقتبس يحدد النصاب بشكل مباشر وهو مقدار محدد من المال يجب على المسلم دفعه كزكاة إذا وصل ماله إلى هذا المقدار وحال عليه الحول.
> *al-nass al-muqtabas yuhaddid al-nisab bi-shakl mubashir wa huwa miqdar muhaddad min al-mal yajib 'ala al-muslim daf'uhu ka-zakat idha wasala maluhu ila hadha al-miqdar wa hala 'alayhi al-hawl.*

Failure mode: indirect support not recognized. For a question asking whether a person's Islam is valid if they testify the shahada but believe a wali can act in the universe with God's permission, one citation (a hadith on the conditions for accepting someone's Islam) is labeled non-relevant, though gold marks it relevant. The model's own step-three reasoning states the citation "emphasizes that the two testimonies of faith must be sincere to God alone, without associating anyone else with any attribute of divinity... the text does not directly answer the question, but it offers support for the logic of the answer."

> **Question:** هل يصحّ إسلام شخص ينطق بالشهادتين لكنه يعتقد أن أحد الأولياء يتصرف في الكون بإذن من الله؟ ولماذا؟
> *hal yasihh islam shakhsin yantiq bil-shahadatayn lakinnahu ya'taqid anna ahad al-awliya' yatasarraf fi al-kawn bi-idhn min Allah? wa limadha?*
> **Citation:** أمرت أن أقاتل الناس حتى يشهدوا أن لا إله إلا الله وأن محمدًا رسول الله، ويقيموا الصلاة، ويؤتوا الزكاة، فإذا فعلوا ذلك عصموا مني دماءهم وأموالهم إلا بحق الإسلام، وحسابهم على الله
> *umirtu an uqatila al-nasa hatta yashhadu an la ilaha illa Allah wa anna Muhammadan rasul Allah, wa yuqimu al-salata, wa yu'tu al-zakata, fa-idha fa'alu dhalika 'asamu minni dima'ahum wa amwalahum illa bi-haqqi al-islam, wa hisabuhum 'ala Allah*
> **Reasoning:** النص المقتبس لا يجيب بشكل مباشر على السؤال حول صحة إسلام شخص يعتقد أن أحد الأولياء يتصرف في الكون بإذن من الله، ولكنه يوضح شروط الإسلام الأساسية.
> *al-nass al-muqtabas la yujib bi-shakl mubashir 'ala al-su'al hawla sihhat islam shakhsin ya'taqid anna ahad al-awliya' yatasarraf fi al-kawn bi-idhn min Allah, wa lakinnahu yuwaddih shurut al-islam al-asasiyya.*

This is a description of indirect support under the task's own Case 2 definition, stated in the model's own words, yet the final label does not reflect it. Across every citation retrieved for this question in the log, the system predicts non-relevant throughout, a complete miss on a question where gold marks the citation as relevant.

Failure mode: citation hallucination within the reasoning trace. For the question asking whether memorization is better than understanding in learning the Qur'an, Fanar-S-1-7B's chain of thought repeatedly reasons about a specific, well-known hadith on the virtue of learning and teaching the Qur'an, even when the citation actually provided for that row is a different, unrelated verse or hadith. In one representative case, the provided citation is a short verse on proclaiming God's favor, unrelated to memorization or understanding, yet the model's own step two states "the Prophet Muhammad, peace be upon him, encouraged memorizing and understanding the Qur'an, as stated in the noble hadith: 'the best of you is the one who learns the Qur'an and teaches it' (narrated by Bukhari and Muslim)." No such hadith was provided as the citation for this row.

> **Question:** هل الحفظ أفضل من الفهم في تعليم القرآن؟
> *hal al-hifz afdal min al-fahm fi ta'lim al-qur'an?*
> **Citation:** وأما بنعمة ربك فحدث
> *wa amma bi-ni'mati rabbika fahaddith*
> **Reasoning:** النبي محمد صلى الله عليه وسلم شجع على حفظ القرآن وفهمه، كما جاء في الحديث الشريف: 'خيركم من تعلم القرآن وعلمه' (رواه البخاري ومسلم).
> *al-nabi Muhammad salla Allahu 'alayhi wa sallam shajja'a 'ala hifz al-qur'an wa fahmihi, kama ja'a fi al-hadith al-sharif: 'khayrukum man ta'allama al-qur'ana wa 'allamahu' (rawahu al-bukhari wa muslim).*

The model then bases its reasoning on this introduced hadith rather than on the provided citation, concluding that it is a direct answer and predicting it as relevant, while the gold label is non-relevant. This pattern occurs in seven of the citations retrieved for this question, and in six of those seven cases the resulting prediction is incorrect. The same question, evaluated with an identical prompt using ALLaM-7B, shows no instance of this pattern in the same citation set. We checked two possible prompt-level explanations directly. The specific hadith does not appear in the static prompt template and is also absent from the dynamically sampled few-shot demonstrations used for this run, which were drawn with a fixed random seed and were therefore identical across

both backends. These observations show that, in these instances, Fanar-S-1-7B based its reasoning on a hadith that was topically associated with the question but was not present in the citation provided to the model.

A category the normalization logic does not recognize. The chain of thought prompt requires one of four specific category labels. In several rows, the raw category output does not match any of them. For the question asking about the nisab threshold for zakat, the provided citation states the threshold directly (five awaq of silver). The model's reasoning recognizes its relevance to the nisab threshold, although it incorrectly describes the citation as part of a Qur'anic verse.

> **Question:** ما هو نصاب الزكاة
> *ma huwa nisab al-zakat*
> **Citation:** ليس فيما دون خمس أواق صدقة
> *laysa fima duna khamsi awaqin sadaqa*
> **Reasoning:** النص المقتبس يشير إلى جزء من الآية القرآنية التي تحدد النصاب للزكاة، ولكنه لا يقدم معلومات مباشرة حول نصاب الزكاة.
> *al-nass al-muqtabas yushir ila juz'in min al-aya al-qur'aniyya allati tuhaddid al-nisab lil-zakat, wa lakinnahu la yuqaddim ma'lumat mubashira hawla nisab al-zakat.*

The raw category output for this row is the Arabic word for indirect, "غير مباشرة" (*ghayr mubashira*), not one of the four expected English labels, and the normalization logic maps it to unrecognized rather than to any of the four categories. A final relevant label of False is still produced despite this, though gold marks the citation relevant. This is a distinct failure from a parse error: the model did produce a category, just not one the normalization step was built to recognize.

A parse failure, found rather than assumed absent. One row in the log has every chain of thought field (question intent, citation content, reasoning, raw category) recorded as missing, with an explicit parse error flag set, yet the system still returns a relevant prediction, which happens to match gold on this specific row. Despite the parse failure, the model produced the correct label without any reasoning being recorded in the log.